# CHARLES WANG


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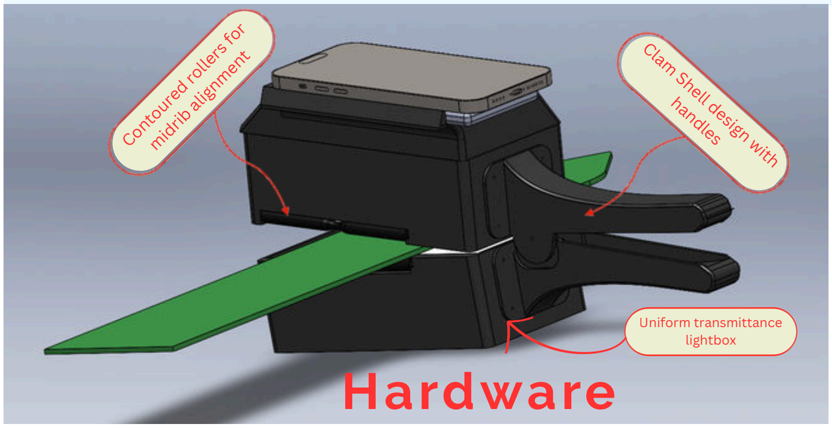
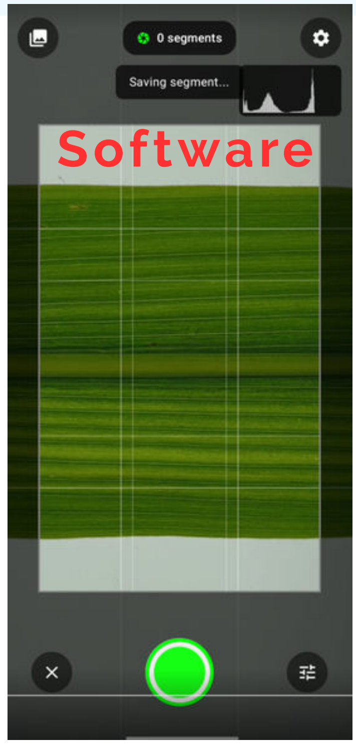
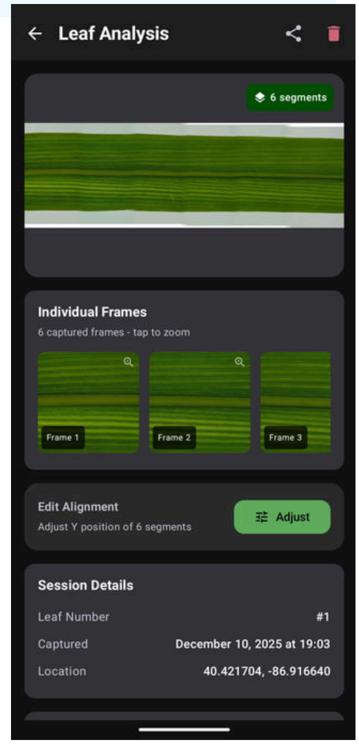
## SMARTPHONE PLANT IMAGING SYSTEM WITH AI-POWERED DIAGNOSIS - LEAFDOC

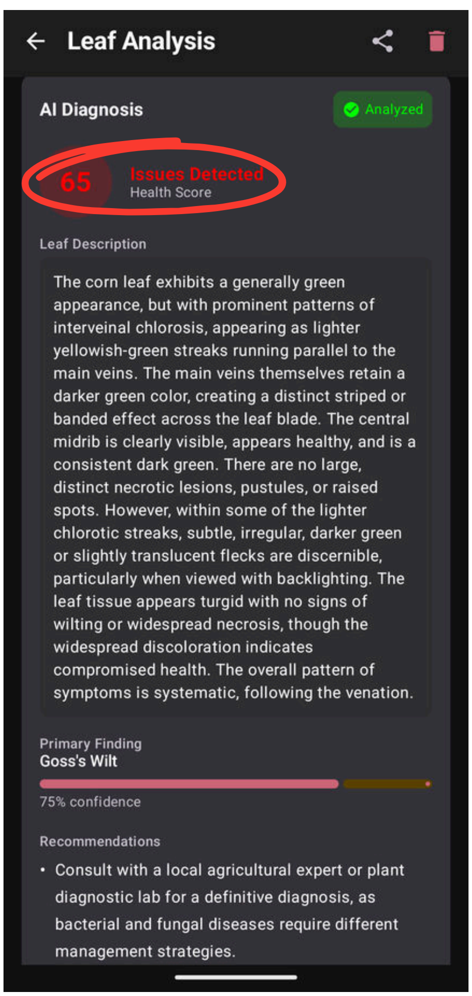



**What?**

- **$50** handheld lightbox smartphone attachment that blocks ambient light and provides uniform LED illumination
- Contoured rollers **flatten** and center the sample on entry, ensuring repeatable positioning without manual adjustment
- Lightbox blocks external noise and provides **uniform** illumination
- **MagSafe** mounting locks smartphone above the leaf in a fixed, repeatable position






**How?**

- Modeled and iterated through 7 enclosure revisions in **SolidWorks**, 3D printing each to test light leak, roller clearance, and component fit
- Wrote an alignment stitching algorithm that assembles overlapping frames into a single full-length image
- Built a native **Android app** (Kotlin, CameraX) with manual camera controls and a multi-provider AI diagnostic pipeline

**Results**

- User captures image using smartphone then can organize and manage on app
- Cloud based AI delivers a **health score**, primary finding, and confidence level in-app, no export or external tools needed

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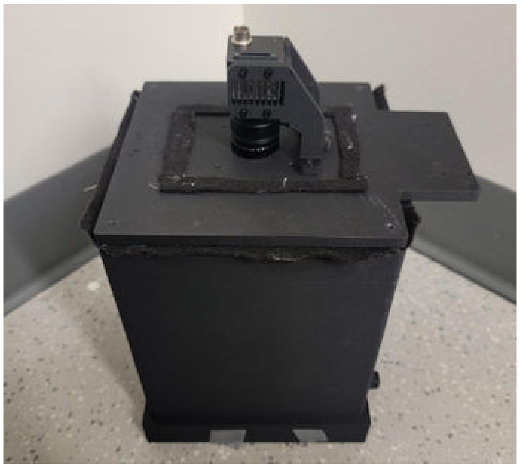
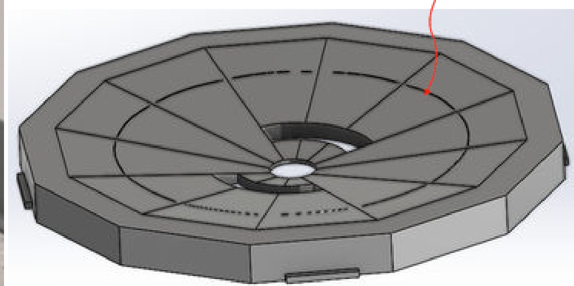
## PORTABLE PHOTOSYNTHETIC EFFICIENCY IMAGING SYSTEM - FLUROSCAN



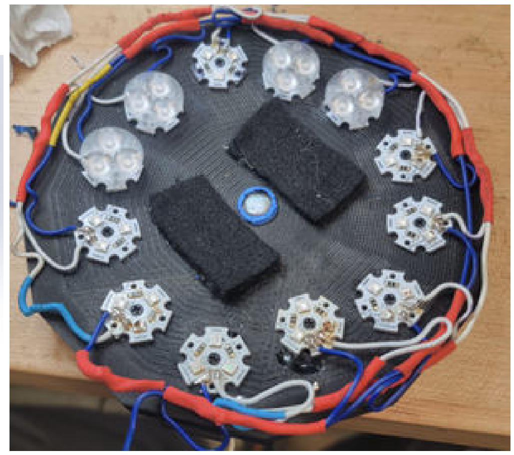
### What?

- Direct measurement of photosynthetic efficiency
- Real-time non-invasive imaging with **early detection** of plant stress
- More affordable and accessible technology for the masses



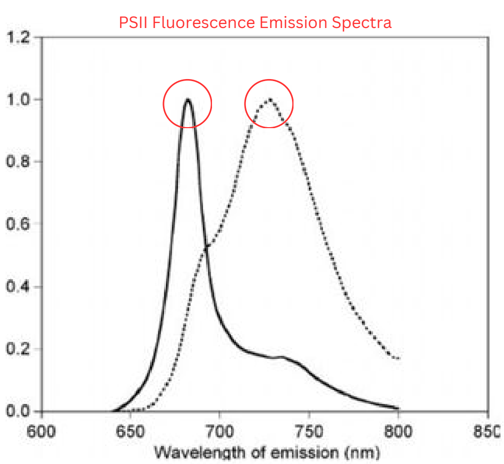

### How?

- Used **Solidworks** to prototype
- Used **Raspberry Pi** for camera and system control
- Used **Bambu Labs** to 3D print
- Dark adapted leaves to fully open PSII
- Saturated photosystems with 450 nm LEDs and captured images through a 680/720 nm filter



### Specifications

- Achieved 450 nm excitation light at 4000 μmol m−2s−1 using 12 LEDs in 3-up configuration

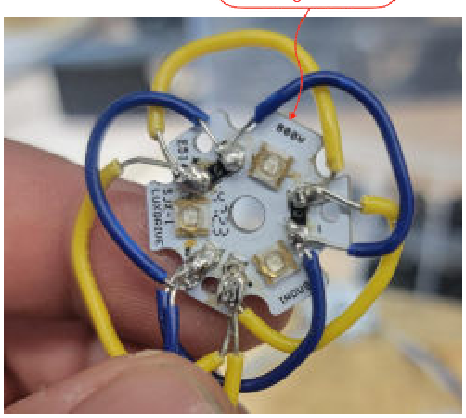



### Theory

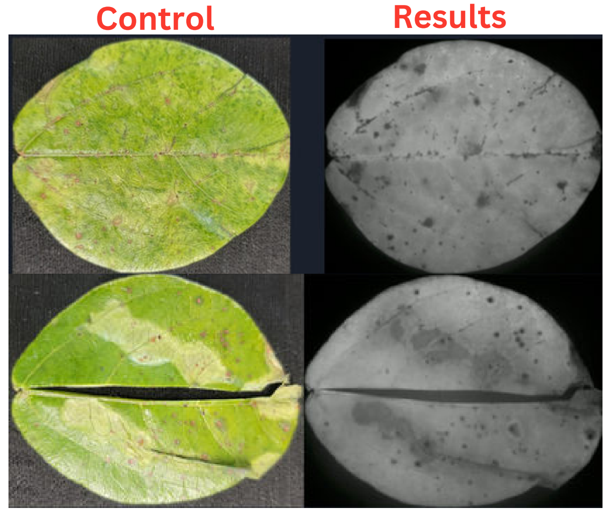
- A weak, non-actinic light will excite and open PSII reaction centers and give the **baseline fluorescence** (Fo)
- A strong pulse of light will close all PSII reaction centers and give **maximum fluorescence** (fm)
- **Ratio** represents maximum efficiency of PSII energy conversion

$$Fv = Fm - Fo$$

$$Fv/Fm$$




**Control** **Results**



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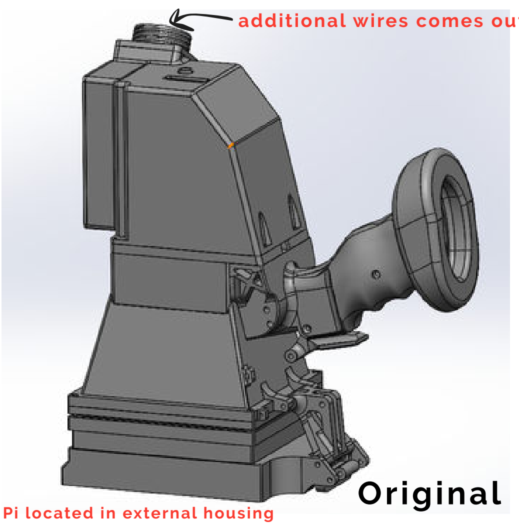
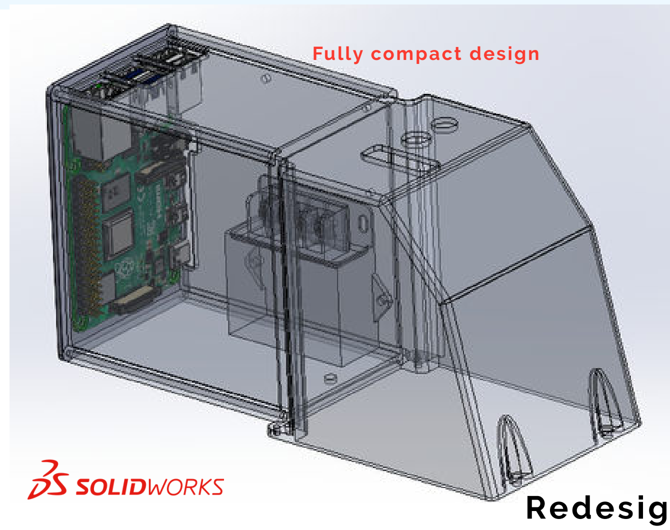
## MULTISPECTRAL HANDHELD IMAGING DEVICE




**What?**

- Plant phenotyping device that collects **high resolution images** (0.04mm/pixel)
- Redesigned enclosure houses all electronics internally, eliminating external wiring for a fully field-portable system

**How?**

- Used **Solidworks** to prototype
- Used **Bambu Labs** to 3D print
- Used **Raspberry Pi** for camera and system control
- Reverse-engineered and documented all subsystems
- Soldered 120+ LEDs across 6 wavelength bands with MOSFET transistors and 12V/5V power conversion

**Results**

- **High signal to noise ratio** with ambient light blocked
- Cleary captured the secondary veins on leaves
- Detected plant stress days before visual symptoms
- Enabled co-researcher's ML model to achieve 90% classification accuracy in a peer-reviewed publication (co-author)